

Practical Group Challenge

Synthetic Data Generation for Economic Policymaking | IndabaX Ghana 2026

Group name / number:

Members:

Your Challenge

Your group will use the practical notebook to answer a policy question that is **similar to the demonstration**, but not identical to it.

The demonstration asks whether DHS child stunting differs by synthetic poverty status. Your task is to choose a **different DHS child outcome**, run the same style of analysis, and decide whether the synthetic-data evidence is credible enough to use.

Prizes: the top two group submissions will receive prizes.

Submission

Submit one group memo by email.

- **Send to:** augidenteh@gmail.com
- **Subject line:** IndabaX synthetic data challenge - Group [your group name/number]
- **Attach or paste:** a one-page memo, or a knitted HTML/PDF excerpt from your notebook
- **Include:** group name/number and member names
- **Deadline:** the deadline announced in class

Outcome Choices

Choose **one** DHS child outcome other than **stunted**:

Outcome variable	Interpretation
underweight	child is underweight
wasted	child is wasted
diarrhea_recent	child had diarrhea recently
fever_recent	child had fever recently
cough_recent	child had cough recently

Outcome variable	Interpretation
dpt3	child received DPT3 vaccination
measles	child received measles vaccination

What To Do

1. **State a policy question.** It should name both the DHS outcome and the synthetic welfare target variable.
2. **Run the analysis for your chosen outcome.** Modify the child-outcome section of the notebook by replacing the demonstration outcome (**stunted**) with your chosen variable.
3. **Report the synthetic-poor vs synthetic-nonpoor gradient.** Use weighted rates and compare the result across imputed datasets. Do not report only one imputed dataset.
4. **Bring validation evidence.** Use at least two pieces of evidence from the notebook: GLSS holdout recovery, the DHS transport check, subgroup gaps, or stability across imputed datasets.
5. **Make a verdict.** Choose one: credible, useful but limited, or not credible. Then explain what evidence supports the verdict.

Memo Template

Use this structure in your submitted memo (not necessarily in this table)

Memo item
Group name / members
Policy question
DHS child outcome
Synthetic welfare measure
Synthetic-poor rate
Synthetic-nonpoor rate
Gradient
Validation evidence 1
Validation evidence 2
Verdict: credible / useful but limited / not credible